Database Management System (DBMS)

Previous Year Question Paper – 2023
Time: 3 Hours Maximum Marks: 100

Note: This is a sample/dummy placeholder question paper generated for demonstration purposes only. It does not represent an official examination paper. Some questions repeat across years with different wording, as is common in real previous-year question compilations.

Section A – Short Answer Questions (Answer any 5)

Q1. Define a database transaction. Draw and explain the transaction state diagram. *(5 marks each)*

Q2. What is Boyce-Codd Normal Form (BCNF)? How is it different from 3NF? *(5 marks each)*

Q3. List and explain any four aggregate functions used in SQL. *(5 marks each)*

Q4. Define a database trigger. Explain its use with an example. *(5 marks each)*

Q5. What is a serializable schedule? Explain its significance in concurrent execution. *(5 marks each)*

Section B – Medium Answer Questions (Answer any 4)

Q1. Describe one-to-one, one-to-many and many-to-many relationships with suitable examples. *(10 marks each)*

Q2. Define a stored procedure. Write a simple example and explain its advantages. *(10 marks each)*

Q3. What is Two-Phase Locking (2PL)? Explain its growing and shrinking phases. *(10 marks each)*

Q4. What is a data warehouse? How does it differ from a traditional database? *(10 marks each)*

Section C – Long Answer Questions (Answer any 3)

Q1. Explain the need for concurrency control in DBMS and describe lock-based and timestamp-based protocols. *(15 marks each)*

Q2. Model an Online Shopping System with an ER diagram, including Customer, Product, Order and Payment entities, and derive the relational schema. *(15 marks each)*

Q3. Take an unnormalized relation and normalize it step by step up to BCNF, explaining each step. *(15 marks each)*

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